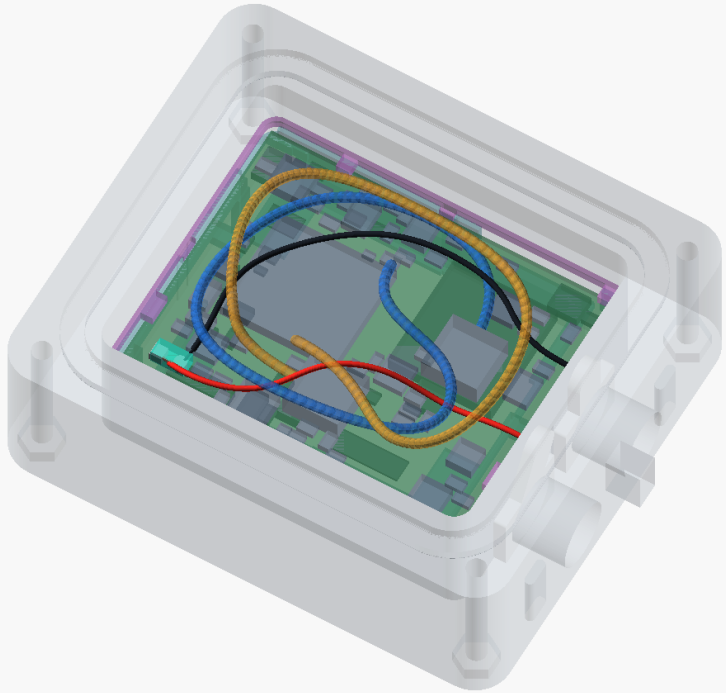
CT-INTERNAL-CHARGE-003-A / R19

PROTOTYPE REVIEW - manufacturing and physical qualification HOLD



DISPENSING ENVELOPES / mm

pcb-pair

X: -17.200 to -13.400
Y: -13.050 to -11.120
Z: 11.241 to 12.950

red-terminal

X: 20.700 to 22.450
Y: -3.950 to -2.650
Z: 12.800 to 14.250

black-terminal

X: 20.700 to 22.450
Y: 5.350 to 6.650
Z: 12.800 to 14.250

Dimensions use the assembled enclosure frame. Total placement/service allowance +/-0.05mm.

The PCB owns one shared insulating block. Each metal terminal owns a separate block.

PTFE adhesion is not relied on. Inspect both solder joints before covering them.

27/27 static poses pass; 52 added checks per pose. Minimum component gap: 0.22mm.

The geometry does not establish bond strength, continuous motion or a production process.

Factory trial: lid open, battery absent, 21..25C / 40..60% RH / >=168h plus passing witnesses.

Handling qualification target: 0.5N for 60s, anchor movement <=0.03mm. Not yet tested.

Use restraint.json and README.md for covered lengths, substrate tests and release requirements.

CT-INTERNAL-CHARGE-003-A / R19

PROTOTYPE REVIEW - manufacturing and physical qualification HOLD





RED FORMING NEST

Overall: 47.39 x 18.33 x 9.00mm

Centerline reference: 43.7658mm

Bounded tool clearance: 0.0730mm

BLACK FORMING NEST

Overall: 44.77 x 30.33 x 9.00mm

Centerline reference: 54.9209mm

Bounded tool clearance: 0.0459mm

The tools sit on a bench and are removed before final assembly. They do not fit inside the collar.

Channels are open vertically. Lift the wire straight out, then inspect released shape and springback.

Maximum wire OD 0.6604mm; tool tolerance +/-0.05mm. Tool material/process must meet that tolerance.

Strip 1.8 +/-0.1mm at the terminal and 2.4 +/-0.1mm at the PCB. Verify actual ends in assembly.

The recorded curve lengths are not a qualified cut-length tolerance. Do not stretch wires to fit.

Confirm lid installation, adhesive deformation and cyclic motion on samples before release.